DR. A.P.J. ABDUL KALAM TECHNICAL UNIVERSITY UTTAR PRADESH, LUCKNOW

EVALUATION SCHEME & SYLLABUS
(*Second Year*)

for

Bachelor of Technology- Data Science

(Effective from the Session: 2025-26)

Semester-III

SN	Subject Code	Subject	Type	Category	Periods			Sessional Component		Sessional (SW) (TS/P)	End Semester Examination (ESE)	Total SW+ ESE	Credit Cr
					L	T	P	CT	TA	CT+TA	TE/PE		
1	BOE3**/ BAS303	Science Based Open Elective/BSC (Math-III/Math IV/ Math V)	T	ES/BS	3	1	0	20	10	30	70	100	4
2	BVE301/ BAS301	Universal Human Value and Professional Ethics/ Technical Communication	T	VA/HS	2	1	0	20	10	30	70	100	3
3	BCS301	Data Structure	T	PC	3	1	0	20	10	30	70	100	4
4	BCS302	Computer Organization and Architecture	T	PC	3	1	0	20	10	30	70	100	4
5	BDS301	Introduction to Data Science	T	PC	2	1	0	20	10	30	70	100	3
6	BCS351	Data Structure Lab	P	PC	0	0	2		50	50	50	100	1
7	BCS352	Computer Organization and Architecture Lab	P	PC	0	0	2		50	50	50	100	1
8	BCS353	Web Designing Workshop	P	PC	0	0	2		50	50	50	100	1
10	BCC301/ BCC302	Cyber Security/Python programming	T	VA	2	0	0	20	10	30	70	100	2
11	BCC351	Internship Assessment / Mini Project*	P							100		100	2
		Total			**15**	**5**	**6**						**25**

- **Mathematics –III** for CE / ENV and allied branches
- **Mathematics-IV** for Computer/Electronics/Electrical & allied Branches, Mechanical & Allied Branches Textile/Chemical & allied Branches
- **Mathematics-V** for Bio Technology / Agriculture Engineering

Semester-IV

SN	Subject Code	Subject	Type	Category	Periods			Sessional Component		Sessional (SW)	End Semester Examination (ESE)	Total SW+ESE	Credit Cr
					L	T	P	CT	TA	CT+TA	TE/PE		
1	BOE4**/ BAS403	Science Based Open Elective /BSC (Math-III/ Math IV/ Math V)	T	BS/ ES	3	1	0	20	10	30	70	100	4
2	BAS401/ BVE401	Technical Communication / Universal Human Value and Professional Ethics	T	HS/ VA	2	1	0	20	10	30	70	100	3
3	BCS401	Operating System	T	PC	3	1	0	20	10	30	70	100	4
4	BDS401	Foundation of Artificial Intelligence and Machine Learning	T	PC	3	1	0	20	10	30	70	100	4
5	BCS403	Object Oriented Programming with Java	T	PC	2	1	0	20	10	30	70	100	3
6	BCS451	Operating System Lab	P	PC	0	0	2		50	50	50	100	1
7	BCS452	Object Oriented Programming with Java Lab	P	PC	0	0	2		50	50	50	100	1
8	BDS451	Artificial Intelligence & Machine Learning Lab	P	PC	0	0	2		50	50	50	100	1
9	BCC401/ BCC402	Cyber Security / Python Programming	P	VA	2	0	0	20	10	30	70	100	2
10	BVE451/ BVE452	Sports and Yoga - II/NSS-II	P	VA	0	0	3			100		100	0
		Total			**15**	**5**	**9**						**23**

*The Mini Project or internship (4 weeks) will be done during summer break after 4th Semester and will be assessed during V semester.

SYLLABUS

BCS301	DATA STRUCTURE	
Course Outcome (CO)		**Bloom's Knowledge Level (KL)**
At the end of course, the student will be able to understand		
CO1	Explain the concepts of Abstract Data Types (ADT), algorithm analysis, asymptotic notations, and the representation of arrays and linked lists for efficient data organization.	**K_2**
CO2	Apply stacks, queues, and recursion techniques to solve computational problems using appropriate data structures and algorithms.	**K_3**
CO3	Analyze the performance of searching, hashing, and sorting algorithms based on their time and space complexities for different problem scenarios.	**K_4**
CO4	Analyze tree-based data structures and implement tree traversal, searching, insertion, and deletion algorithms for efficient data management.	**K_4**
CO5	Evaluate graph representations, graph traversal algorithms, minimum spanning tree algorithms, and shortest path algorithms to solve real-world problems efficiently	**K_5**
DETAILED SYLLABUS		**3-1-0**

Unit	Topic	Proposed Lecture
I	**Introduction:** Basic Terminology, Elementary Data Organization, Built in Data Types in C. Algorithm, Efficiency of an Algorithm, Time and Space Complexity, Asymptotic notations: Big Oh, Big Theta and Big Omega, Time-Space trade-off. Abstract Data Types (ADT) **Arrays**: Definition, Single and Multidimensional Arrays, Representation of Arrays: Row Major Order, and Column Major Order, Derivation of Index Formulae for 1-D,2-D, 3-D and n-D Array Application of arrays, Sparse Matrices and their representations. **Linked lists:** Array Implementation and Pointer Implementation of Singly Linked Lists, Doubly Linked List, Circularly Linked List, Operations on a Linked List. Insertion, Deletion, Traversal, Polynomial Representation and Addition Subtraction & Multiplications of Single variable & Two variables Polynomial.	**08**
II	**Stacks:** Abstract Data Type, Primitive Stack operations: Push & Pop, Array and Linked Implementation of Stack in C, Application of stack: Prefix and Postfix Expressions, Evaluation of postfix expression, Iteration and Recursion- Principles of recursion, Tail recursion, Removal of recursion Problem solving using iteration and recursion with examples such as binary search, Fibonacci numbers, and Hanoi towers. Tradeoffs between iteration and recursion. **Queues:** Operations on Queue: Create, Add, Delete, Full and Empty, Circular queues, Array and linked implementation of queues in C, Dequeue and Priority Queue.	**08**

III	**Searching:** Concept of Searching, Sequential search, Index Sequential Search, Binary Search. Concept of Hashing & Collision resolution Techniques used in Hashing. **Sorting:** Insertion Sort, Selection, Bubble Sort, Quick Sort, Merge Sort, Heap Sort and Radix Sort.	**08**

IV	**Trees:** Basic terminology used with Tree, Binary Trees, Binary Tree Representation: Array Representation and Pointer (Linked List) Representation, Binary Search Tree, Strictly Binary Tree, Complete Binary Tree. A Extended Binary Trees, Tree Traversal algorithms: Inorder, Preorder and Postorder, Constructing Binary Tree from given Tree Traversal, Operation of Insertion, Deletion, Searching & Modification of data in Binary Search. Threaded Binary trees, Traversing Threaded Binary trees. Huffman coding using Binary Tree. Concept & Basic Operations for AVL Tree, B Tree & Binary Heaps	**08**
V	**Graphs:** Terminology used with Graph, Data Structure for Graph Representations: Adjacency Matrices, Adjacency List, Adjacency. Graph Traversal: Depth First Search and Breadth First Search, Connected Component, Spanning Trees, Minimum Cost Spanning Trees: Prims and Kruskal algorithm. Transitive Closure and Shortest Path algorithm: Warshal Algorithm and Dijikstra Algorithm.	**08**

Text books:

1. Aaron M. Tenenbaum, Yedidyah Langsam and Moshe J. Augenstein, "Data Structures Using C and C++", PHI Learning Private Limited, Delhi India.
2. Gilberg ,Forouzan, Data Structures: A Pseudocode Approach with C 3rd edition , Cengage Learning publication.
3. Horowitz and Sahani, "Fundamentals of Data Structures", Galgotia Publications Pvt Ltd Delhi India.
4. Lipschutz, "Data Structures" Schaum's Outline Series, Tata McGraw-hill Education (India) Pvt. Ltd.
5. Thareja, "Data Structure Using C" Oxford Higher Education.
6. AK Sharma, "Data Structure Using C", Pearson Education India.
7. Rajesh K. Shukla, "Data Structure Using C and C++" Wiley Dreamtech Publication.
8. Michael T. Goodrich, Roberto Tamassia, David M. Mount "Data Structures and Algorithms in C++", Wiley India.
9. P. S. Deshpandey, "C and Data structure", Wiley Dreamtech Publication.
10. R. Kruse etal, "Data Structures and Program Design in C", Pearson Education.
11. Berztiss, AT: Data structures, Theory and Practice, Academic Press.
12. Jean Paul Trembley and Paul G. Sorenson, "An Introduction to Data Structures with applications", McGraw Hill.
13. Adam Drozdek "Data Structures and Algorithm in Java", Cengage Learning

BCS302	COMPUTER ORGANIZATION AND ARCHITECTURE	
	Course Outcome (CO)	**Bloom's Knowledge Level (KL)**
	At the end of course, the student will be able to understand	
CO1	Explain the organization of a digital computer system, processor architecture, register organization, buses, memory transfer, and addressing modes.	K_2
CO2	Analyze arithmetic and logic unit (ALU) operations.	K_4
CO3	Analyze the design and operation of hardwired and microprogrammed control units, instruction execution cycles, and pipelining techniques for processor performance enhancement.	K_4
CO4	Evaluate the performance of memory hierarchy, cache memory, virtual memory, and secondary storage techniques for efficient memory management.	K_5
CO5	Evaluate input/output organization, interrupt handling, data transfer techniques, and serial communication interfaces for effective interaction between the processor and peripheral devices.	K_5

	DETAILED SYLLABUS	**3-1-0**
Unit	**Topic**	**Proposed Lecture**
I	**Introduction**: Functional units of digital system and their interconnections, buses, bus architecture, types of buses and bus arbitration. Register, bus and memory transfer. Processor organization, general registers organization, stack organization and addressing modes.	**08**
II	**Arithmetic and logic unit:** Look ahead carries adders. Multiplication: Signed operand multiplication, Booths algorithm and array multiplier, Division and logic operations, Floating point arithmetic operation, Arithmetic & logic unit design. IEEE Standard for Floating Point Numbers	**08**
III	**Control Unit:** Instruction types, formats, instruction cycles and sub cycles (fetch and execute etc), microoperations, execution of a complete instruction. Program Control, Reduced Instruction Set Computer, Pipelining. Hardwire and micro programmed control: micro programme sequencing, concept of horizontal and vertical microprogramming.	**08**
IV	**Memory:** Basic concept and hierarchy, semiconductor RAM memories, 2D & 2 1/2D memory organization. ROM memories. Cache memories: concept and design issues & performance, address mapping and replacement Auxiliary memories: magnetic disk, magnetic tape and optical disks Virtual memory: concept implementation.	**08**
V	**Input / Output**: Peripheral devices, I/O interface, I/O ports, Interrupts: interrupt hardware, types of interrupts and exceptions. Modes of Data Transfer: Programmed I/O, interrupt initiated I/O and Direct Memory Access., I/O channels and processors. Serial Communication: Synchronous & asynchronous communication, standard communication interfaces.	**08**

Text books:

1. Computer System Architecture - M. Mano
2. Carl Hamacher, Zvonko Vranesic, Safwat Zaky Computer Organization, McGraw-Hill, Fifth Edition, Reprint 2012
3. John P. Hayes, Computer Architecture and Organization, Tata McGraw Hill, Third Edition, 1998. Reference books
4. William Stallings, Computer Organization and Architecture-Designing for Performance, Pearson Education, Seventh edition, 2006.
5. Behrooz Parahami, "Computer Architecture", Oxford University Press, Eighth Impression, 2011.
6. David A. Patterson and John L. Hennessy, "Computer Architecture-A Quantitative Approach", Elsevier, a division of reed India Private Limited, Fifth edition, 2012
7. Structured Computer Organization, Tannenbaum(PHI)

BDS301: Introduction to Data Science		
Course Outcome (CO)		**Bloom's Knowledge Level (KL)**
At the end of course, the student will be able to understand		
CO1	Explain the fundamental concepts of Data Science, its life cycle, data types, and applications in solving real-world problems.	K_2
CO2	Apply statistical methods and exploratory data analysis techniques to summarize, analyze, and interpret datasets for informed decision-making.	K_3
CO3	Develop Python programs using data science libraries to perform data analysis and implement the data science project development process.	K_6
CO4	Analyze and preprocess datasets using data wrangling techniques and create effective visualizations for data interpretation.	K_6
CO5	Evaluate the use of modern data science tools, ethical principles, AI agents, and their applications in real-world scenarios.	K_5
DETAILED SYLLABUS		
Unit	**Topic**	**Proposed Lecture**
I	**Introduction to Data Science:** Introduction to Data Science Definition, scope, and evolution of Data Science, Data Science Life Cycle, Data Analytics and Business Intelligence. Real-world applications, Overview of Big Data and its challenges. Structured vs. Unstructured Data, Data types, Overview of datasets, Data Collection, Preprocessing, Analysis, Visualization, and Model Building.	**08**
II	**Exploratory Data Analysis:** Measures of Central Tendency, Measures of Dispersion. Scatter diagram, covariance, Correlation, Statistical Analysis, Hypothesis testing: t -Test, chi- square Test, F-Test, Z-Test, ANOVA	**08**
III	**Programming in Data Science:** Introduction to Python for Data Science Basic Syntax, Data Types, and Operators. Control Structure, Functions and Modules, Working with Data Science Libraries NumPy, Pandas, Scikit-Learn. Project Development Process, Problem Identification and Data Collection, Data Cleaning and Exploration, Model Selection and Implementation, Result Interpretation and Presentation.	**08**
IV	**Data Wrangling, Prepossessing & Visualization (8)** Data Cleaning & Preprocessing Handling missing values, removing duplicates, and outliers, Data transformation, Data Manipulation in Python, Working with NumPy and Pandas, Filtering, Sorting, Grouping, and Aggregation. Data Visualization Techniques Basics of Matplotlib and Seaborn, Types of Graphs: Bar Charts, Histograms, Box Plots, Scatter Plots, Line Graphs, Heatmaps, and Pair Plots.	**08**
V	**Data Science Tools, Ethics and Applications (8)** Tools for Data Science: Introduction, Basics of Tableu, Basics of Power BI, Basics of R-Programming, Applications Ethics in Data Science, AI Agents, types and applications.	**08**

Suggested Readings:

1. "Python for Data Analysis: Data Wrangling with Pandas, NumPy, and IPython" by Wes McKinney (O'Reilly Media)
2. Data Science from Scratch: First Principles with Python" by Joel Grus (O'Reilly Media).
3. Cathy O'Neil and Rachel Schutt. Doing Data Science, Straight Talk from The Frontline. O'Reilly. 2014.
4. W. H. Inmon, "Building the Data Warehouse", 3rd edition.
5. Anahory and Murray. Data warehousing in the real world, Pearson Education/Addison Wesley.
6. Margaret Dunham, Data Mining: Introductory and Advanced Topics, Published by Prentice Hall.
7. Jiawei Han, Micheline Kamber, "Data Mining: Concepts and Techniques",
8. Morgan Kaufmann Publishers, 2002. (www.cs.sfu.ca/~han/DMbook.html).
Alex Berson, Stephen J. Smith, "Data Warehousing, Data Mining, & OLAP", Tata Mcgraw- Hill, 2004.

BCS351- Data Structure Lab

List of Experiments (Indicative & not limited to)

1. **Implementing Sorting Techniques:** Bubble Sort, Insertion Sort, Selection Sort, Shell , Sort, Radix Sort, Quick sort
2. **Implementing Searching and Hashing Techniques:** Linear search, Binary search, Methods for Hashing: Modulo Division, Digit Extraction, Fold shift, Fold Boundary, Linear Probe for Collision Resolution. Direct and Subtraction hashing
3. **Implementing Stacks:** Array implementation, Linked List implementation, Evaluation of postfix expression and balancing of parenthesis, Conversion of infix notation to postfix notation
4. **Implementing Queue:** Linked List implementation of ordinary queue, Array implementation of circular queue, Linked List implementation of priority queue, Double ended queue
5. **Implementing Linked List:** Singly Linked Lists, Circular Linked List, Doubly Linked Lists: Insert, Display, Delete, Search, Count, Reverse (SLL), Polynomial, Addition, Comparative study of arrays and linked list
6. **Implementing Trees:** Binary search tree: Create, Recursive traversal: preorder, post order, in order, Search Largest, Node, Smallest Node, Count number of nodes, Heap: Min Heap, Max Heap: reheap Up, reheap Down, Delete, Expression Tree, Heapsort
7. **Implementing Graphs**: Represent a graph using the Adjacency Matrix, BFS, Find the minimum spanning tree (using any method Kruskal's Algorithm or Prim's Algorithm) Self Learning Topics: Shortest Path Algorithm

BCS352- Computer Organization Lab

List of Experiments (Indicative & not limited to)

1. Implementing HALF ADDER, FULL ADDER using basic logic gates
2. Implementing Binary -to -Gray, Gray -to -Binary code conversions.
3. Implementing 3-8 line DECODER.
4. Implementing 4x1 and 8x1 MULTIPLEXERS.
5. Verify the excitation tables of various FLIP-FLOPS.
6. Design of an 8-bit Input/ Output system with four 8-bit Internal Registers.
7. Design of an 8-bit ARITHMETIC LOGIC UNIT.
8. Design the data path of a computer from its register transfer language description.
9. Design the control unit of a computer using either hardwiring or microprogramming based on its register transfer language description.
10. Implement a simple instruction set computer with a control unit and a data path.

BCS353- Web Designing Workshop

Syllabus:

- **HTML:** Elements, attributes, heading, paragraph, styles, comments, links, images, favicon, tables, list, class, id, HTML forms, HTML media, navigation bar.
- **CSS**: Types of CSS, colors, background, margins, padding, height, width, text, font, icon, links, list, tables, display, z-index, float, overflow, CSS media queries, inline block, navigation bar, image gallery, forms, round corners
- **BOOTSTRAP** : Fundamentals of implementing responsive web design ,Use Balsamiq to mockup and wireframe websites, The fundamentals of UI design for websites ,How to install the Bootstrap framework ,Understanding the Bootstrap grid layout system, How to use bootstrap containers to layout your website easily, Use other Bootstrap components such as buttons ,Adding symbols using Font Awesome, Bootstrap carousels. Add Bootstrap cards to your website. Using Bootstrap navigation bars,
- **JavaScript** script, function, output, statement, variables, operators, datatypes, objects, events, string methods, Arrays, if else, switch, loop for, loop in, loop for, debugging, validation of forms , Functions and invocation patterns Discussion of ECMAScripts Intermediate JavaScript , JS Expressions, Operators, Statements and Declarations , Object-Oriented Programming JS Objects and Prototypes, `This`, Scope and Closures Objects and Prototypes Refactoring and Debugging

Textbook

1. Meloni, J. C., Kyrnin, J. (2018). HTML, CSS, and JavaScript All in One: Covering HTML5, CSS3, and ES6, Sams Teach Yourself. United Kingdom: Pearson Education.
2. McGrath, M. (2020). HTML, CSS & JavaScript in easy steps. United Kingdom: In Easy Steps Limited.

Reference Books

1. Duckett, J. (2014). Web Design with HTML, CSS, JavaScript and JQuery Set. United Kingdom: Wiley.
2. Fajfar, I. (2015). Start Programming Using HTML, CSS, and JavaScript. United Kingdom: CRC Press.

List of Experiments (Indicative & not limited to)

Experiment No.	List of Experiments (Indicative & not limited to)
1	Designthefollowingstaticwebpagesrequiredforanonlinebookstorewebsite. **HOMEPAGE:** - The static home page must contain three **frames**. - Top frame: Logo and the college name and links to Homepage, Login page, Registration page, Catalogue page and Cart page (the description of these pages will be given below).

For example: When you click the link **"CSE"** the catalogue for **CSE** Books should be displayed in the Right frame. Right frame: The *pages to the links in the left frame must be loaded here*. Initially this page contains description of the web site.

Logo	Web Site Name			
Home	Login	Registration	Catalogue	Cart
CSE ECE EEE CIVIL	Description of the WebSite			

2.

LOGINPAGE:

This page looks like below:

Logo	WebSite Name			
Home	Login	Registration	Catalogue	Cart
CSE ECE EEE CIVIL	Login Page User Name: Passwords: Submit Reset			

3.

CATOLOGUE PAGE: The catalogue page should contain the details of all the books available in the website in a table. The details should contain the following:

1. Snap shot of Cover Page.
2. Author Name.
3. Publisher.
4. Price.
5. Add to cart button.

Logo	WebSite Name			
Home	Login	Registration	Catalogue	Cart
CSE ECE		Book:XMLBible Author : Winston Publication:Wiely	$40.5	Add to cart
EEE CIVIL		Book :AI Author:S.Russel Publication:Princetonhall	$63	Add to cart
				

				Book : Java 2 Author:Watson Publication:BPBpublications Book : HTML in 24 hours Author : Sam Peter Publication:Sampublication	$35.5 $50	Add to cart Add to cart

4.

CARTPAGE: The cart page contains the details about the books which are added to the cart. The cart page should look like this:

Logo	Web Site Name				
Home	Login	Registration		Catalogue	Cart
CSE	Book name	Price	Quantity	Amount	
ECE	Java 2	$35.5	2	$70	
EEE	XML bible	$40.5	1	$40.5	
CIVIL	Total amount	- $130.5			

5.

REGISTRATION PAGE : Create a" registration form" with the following fields

1)Name (Text field)

2)Password (password field)

3) E-mailid(text field)
4) Phone Number(text field)
5) Sex(radio button)
6) Date of birth(3 select boxes)
7) Languages known(checkboxes–English, Telugu, Hindi, Tamil)
8) Address(text area)

6.

Js VALIDATION: Write ***JavaScript*** **to validate** the following fields of the above registration page.

1. Name (Name should contains alphabets and the length should not be less than 6 characters).
2. Password (Password should not be less than 6 characters length).

7.

Js VALIDATION:

3. E-mailid (should not contain any invalid and must follow the standard pattern(name@domain.com)
4. Phone Number(Phone number should contain 10 digits only).

8.

CSS: Design a web page using **CSS(Cascading Style Sheets)** which includes the following:

1) Use different font, styles:
In the style definition you define how each selector should work(font, color etc.). Then, in the body of your pages, you refer to these selectors to activate the styles.

2) Set a background image for both the page and single elements on the page.

9.

CSS:

1) Control the repetition of the image with the background-repeat property.
2) Define styles for links as

	A:link A:visited A:active A:hover
10.	Consider a small topic of your choice on which you can develop static Webpages and try to implement all topics of html, CSS and Js within the topic. Choose any one topic. 1. Your Own Portfolio 2. To-Do List 3. Survey Form 4. A Tribute Page 5. A Questionnaire

FOURTH SEMESTER (DETAILED SYLLABUS)

BCS401	**Operating system**	
Course Outcome (CO)		**Bloom's Knowledge Level (KL)**
At the end of course, the student will be able to understand		
CO1	Explain the architecture, functions, services, and different types of operating systems along with their structural organization.	K_2
CO2	Analyze process management concepts, process synchronization techniques, inter-process communication mechanisms, and classical synchronization problems.	K_4
CO3	Analyze CPU scheduling algorithms, thread management, and deadlock handling techniques for efficient process execution.	K_4
CO4	Evaluate memory management techniques for optimal memory utilization.	K_5
CO5	Evaluate I/O management, disk scheduling algorithms, RAID, and file system organization to ensure efficient storage management and system performance.	K_5

DETAILED SYLLABUS		**3-0-0**
Unit	**Topic**	**Proposed Lecture**
I	**Introduction**: Operating system and functions, Classification of Operating systems- Batch, Interactive, Time sharing, Real Time System, Multiprocessor Systems, Multiuser Systems, Multiprocess Systems, Multithreaded Systems, Operating System Structure- Layered structure, System Components, Operating System services, Reentrant Kernels, Monolithic and Microkernel Systems.	**08**
II	**Concurrent Processes**: Process Concept, Principle of Concurrency, Producer / Consumer Problem, Mutual Exclusion, Critical Section Problem, Dekker's solution, Peterson's solution, Semaphores, Test and Set operation; Classical Problem in Concurrency- Dining Philosopher Problem, Sleeping. Barber Problem; Inter Process Communication models and Schemes, Process generation.	**08**
III	**CPU Scheduling:** Scheduling Concepts, Performance Criteria, Process States, Process Transition Diagram, Schedulers, Process Control Block (PCB), Process address space, Process identification information, Threads and their management, Scheduling Algorithms, Multiprocessor Scheduling. Deadlock: System model, Deadlock characterization, Prevention, Avoidance and detection, Recovery from deadlock.	**08**
IV	**Memory Management:** Basic bare machine, Resident monitor, Multiprogramming with fixed partitions, Multiprogramming with variable partitions, Protection schemes, Paging, Segmentation, Paged segmentation, Virtual memory concepts, Demand paging, Performance of demand paging, Page replacement algorithms, Thrashing, Cache memory organization, Locality of reference.	**08**
V	**I/O Management and Disk Scheduling**: I/O devices, and I/O subsystems, I/O buffering, Disk storage and disk scheduling, RAID. File System: File concept, File organization and access mechanism, File directories, and File	**08**

	sharing, File system implementation issues, File system protection and security.	

Text books:

1. Silberschatz, Galvin and Gagne, “Operating Systems Concepts”, Wiley
2. Sibsankar Halder and Alex A Aravind, “Operating Systems”, Pearson Education
3. Harvey M Dietel, “ An Introduction to Operating System”, Pearson Education
4. D M Dhamdhere, “Operating Systems : A Concept based Approach”, 2nd Edition,
5. TMH 5. William Stallings, “Operating Systems: Internals and Design Principles ”, 6th Edition, Pearson Education

BDS401	Foundation of Artificial and Machine Learning	
	Course Outcome (CO)	**Bloom's Knowledge Level (KL)**
At the end of course, the student will be able to understand		
CO1	Explain the fundamental concepts of Artificial Intelligence, intelligent agents, search techniques, and problem-solving strategies for AI-based applications.	K_2
CO2	Apply first-order logic, knowledge representation techniques, inference mechanisms, and ontological engineering for knowledge-based systems.	K_3
CO3	Analyze the architecture and learning process of Artificial Neural Networks, including perceptron, activation functions, and backpropagation algorithms.	K_4
CO4	Apply supervised machine learning algorithms and evaluate their performance using appropriate classification and validation metrics.	K_3
CO5	Analyze unsupervised learning techniques, clustering algorithms, and dimensionality reduction methods for discovering hidden patterns in data.	K_4

DETAILED SYLLABUS

Unit	Topic	Proposed Lecture
I	**Foundations of Artificial Intelligence** What is AI? History of AI, Foundations of AI. Intelligent Agents: Rational agents, Agent architectures, Environment types. Problem Solving by Search: State-space graphs. Uninformed search strategies, Informed Search: Greedy Best-First Search, A* Search, Hill Climbing	**08**
II	**Knowledge Representation** First order logic – Knowledge Representation, Syntax and semantics in first order logic, Knowledge engineering in first order logic, Inference in First order logic. Prepositional versus first order logic, Unification and lifting, Forward chaining and Backward Chaining, Ontological Engineering.	**08**
III	**Introduction to Artificial Neural Network** Foundations: Biological neurons vs. Artificial neurons, The Perceptron model, limitations of the single-layer perceptron. Network Architecture: Multi-Layer Perceptron (MLP) and Feedforward Neural Networks. Activation Functions: Linear, Sigmoid, Tanh, Softmax, and ReLU. Training the Network-The Backpropagation algorithm, calculating gradients, Loss functions, updating weights using Gradient Descent.	**08**
IV	**Supervised Learning** Introduction to Machine learning, Learning Strategies. Classification Algorithms: Logistic Regression, K-Nearest Neighbors, Decision Trees, Support Vector Machines and Naive Bayes Classifier. Ensemble Learning: Introduction to Bagging and Boosting, Random	**08**

	Forests. Evaluation Metrics: Confusion Matrix, Accuracy, Precision, Recall, F1-Score, ROC-AUC Curve.	
V	**Unsupervised Learning** Supervised vs. Unsupervised Learning, Clustering Algorithms, K-Means Clustering Hierarchical Clustering, DBSCAN etc., Dimensionality Reduction: The Curse of Dimensionality, Introduction to Principal Component Analysis.	**08**

Suggested Readings:

1. Stuart Russell and Peter Norvig, Artificial Intelligence: A Modern Approach, 4th Edition, Pearson, 2021.
2. Elaine Rich and Kevin Knight, Artificial Intelligence, 3rd Edition, McGraw-Hill Education.
3. Tom M. Mitchell, Machine Learning, McGraw-Hill Education, 1997.
4. Ethem Alpaydin, Introduction to Machine Learning, 4th Edition, The MIT Press, Cambridge, Massachusetts, 2020.

BCS403	Object Oriented Programming with Java	
Course Outcome (CO)		**Bloom's Knowledge Level (KL)**
At the end of course, the student will be able to understand		
CO1	Develop the object-oriented programming concepts using Java	K_4
CO2	Implement exception handling, file handling, and multi-threading in Java	K_4
CO3	Apply new java features to build java programs.	K_3
CO4	Analyze java programs with Collection Framework	K_4
CO5	Test web and RESTful Web Services with Spring Boot using Spring Framework concepts	K_5

DETAILED SYLLABUS		**3-1-0**
Unit	**Topic**	**Proposed Lecture**
I	**Introduction:** Why Java, History of Java, JVM, JRE, Java Environment, Java Source File Structure, and Compilation. Fundamental. **Programming Structures in Java:** Defining Classes in Java, Constructors, Methods, Access Specifies, Static Members, Final Members, Comments, Data types, Variables, Operators, Control Flow, Arrays & String. **Object Oriented Programming**: Class, Object, Inheritance Super Class, Sub Class, Overriding, Overloading, Encapsulation, Polymorphism, Abstraction, Interfaces, and Abstract Class. **Packages:** Defining Package, CLASSPATH Setting for Packages, Making JAR Files for Library Packages, Import and Static Import Naming Convention for Packages.	**08**
II	**Exception Handling**: The Idea behind Exception, Exceptions & Errors, Types of Exception, Control Flow in Exceptions, JVM Reaction to Exceptions, Use of try, catch, finally, throw, throws in Exception Handling, In-built and User Defined Exceptions, Checked and Un-Checked Exceptions. **Input /Output Basics**: Byte Streams and Character Streams, Reading and Writing File in Java. **Multithreading:** Thread, Thread Life Cycle, Creating Threads, Thread Priorities, Synchronizing Threads, Inter-thread Communication.	**08**
III	**Java New Features:** Functional Interfaces, Lambda Expression, Method References, Stream API, Default Methods, Static Method, Base64 Encode and Decode, for each Method, Try-with-resources, Type Annotations, Repeating Annotations, Java Module System, Diamond Syntax with Inner Anonymous Class, Local Variable Type Inference, Switch Expressions, Yield Keyword, Text Blocks, Records, Sealed Classes	**08**

IV	**Java Collections Framework:** Collection in Java, Collection Framework in Java, Hierarchy of Collection Framework, Iterator Interface, Collection Interface, List Interface, Array List, Linked List, Vector, Stack, Queue Interface, Set Interface, HashSet, Linked Hash Set, Sorted Set Interface, Tree Set, Map Interface, HashMap Class, Linked Hash Map Class, Tree Map Class, Hash table Class, Sorting, Comparable Interface, Comparator Interface, Properties Class in Java.	**08**
V	**Spring Framework:** Spring Core Basics-Spring Dependency Injection concepts, Spring Inversion of Control, AOP, Bean Scopes- Singleton, Prototype, Request, Session, Application, Web Socket, Auto wiring, Annotations, Life Cycle Call backs, Bean Configuration styles **Spring Boot:** Spring Boot Build Systems, Spring Boot Code Structure, Spring Boot Runners, Logger, BUILDING RESTFUL WEB SERVICES, Rest Controller, Request Mapping, Request Body, Path Variable, Request Parameter, GET, POST, PUT, DELETE APIs, Build Web Applications	**08**

Text Books

1. Herbert Schildt, "Java The complete reference", McGraw Hill Education
2. Craig Walls, “Spring Boot in Action” Manning Publication
3. Steven Holzner, “Java Black Book”, Dreamtech.
4. Balagurusamy E, “Programming in Java”, McGraw Hill
5. Java: A Beginner’s Guide by Herbert Schildt, Oracle Press
6. Greg L. Turnquist “Learning Spring Boot 2.0 - Second Edition”, Packt Publication
7. AJ Henley Jr (Author), Dave Wolf, “Introduction to Java Spring Boot: Learning by Coding”, Independently Published

BCS451- Operating System Lab

List of Experiments (Indicative & not limited to)

1. Study of hardware and software requirements of different operating systems (UNIX, LINUX, WINDOWS XP, WINDOWS7/8
2. Execute various UNIX system calls for
    - i Process management
    - ii File management
    - iii Input/output Systems calls
3. Implement CPU Scheduling Policies:
    - i SJF
    - ii Priority
    - iii FCFS
    - iv Multi-level Queue
4. Implement file storage allocation technique:
    - i Contiguous (using array)

ii Linked –list(using linked-list)
iii Indirect allocation (indexing)

5. Implementation of contiguous allocation techniques:
   i Worst-Fit
   ii Best- Fit
   iii First- Fit
6. Calculation of external and internal fragmentation
   i Free space list of blocks from system
   ii List process file from the system
7. Implementation of compaction for the continually changing memory layout and calculate total movement of data
8. Implementation of resource allocation graph RAG)
9. Implementation of Banker"s algorithm
10. Conversion of resource allocation graph (RAG) to wait for graph (WFG) for each type of method used for storing graph.
11. Implement the solution for Bounded Buffer (producer-consumer) problem using inter process communication techniques-Semaphores
12. Implement the solutions for Readers-Writers problem using inter process communication technique - Semaphore

BCS452- Object Oriented Programming with Java

List of Experiments (Indicative & not limited to)

1. Use Java compiler and eclipse platform to write and execute java program.
2. Creating simple java programs using command line arguments
3. Understand OOP concepts and basics of Java programming.
4. Create Java programs using inheritance and polymorphism.
5. Implement error-handling techniques using exception handling and multithreading.
6. Create java program with the use of java packages.
7. Construct java program using Java I/O package.
8. Create industry-oriented application using Spring Framework.
9. Test RESTful web services using Spring Boot.
10. Test Frontend web application with Spring Boot

BCS453- Cyber Security Workshop

List of Experiments (Indicative & not limited to)

Module 1: Packet Analysis using Wire shark

1. Basic Pack et Inspection: Capture network traffic using Wire shark and analyze basic protocols like HTTP, DNS, and SMTP to understand how data is transmitted and received.
2. Detecting Suspicious Activity: Analyze network traffic to identify suspicious patterns, such as repeated connection attempts or unusual communication between hosts.
3. Malware Traffic Analysis: Analyze captured traffic to identify signs of malware communication, such as command-and-control traffic or data infiltration.
4. Password Sniffing: Simulate a scenario where a password is transmitted in plaintext. Use Wireshark to capture and analyze the packets to demonstrate the vulnerability and the importance of encryption.
5. ARP Poisoning Attack: Set up an ARP poisoning attack using tools like Ettercap. Analyze the captured packets to understand how the attack can lead to a Man-in-the-Middle scenario.

Module 2: Web Application Security using DVWA

1. SQL Injection: Use DVWA to practice SQL injection attacks. Demonstrate how an attacker can manipulate input fields to extract, modify, or delete database information.
2. Cross-Site Scripting (XSS): Exploit XSS vulnerabilities in DVWA to inject malicious scripts into web pages. Show the potential impact of XSS attacks, such as stealing cookies or defacing websites.
3. Cross-Site Request Forgery (CSRF): Set up a CSRF attack in DVWA to demonstrate how attackers can manipulate authenticated users into performing unintended actions.
4. File Inclusion Vulnerabilities: Explore remote and local file inclusion vulnerabilities in DVWA. Show how attackers can include malicious files on a server and execute arbitrary code.
5. Brute-Force and Dictionary Attacks: Use DVWA to simulate login pages and demonstrate brute-force and dictionary attacks against weak passwords. Emphasize the importance of strong password policies.

BDS451	Artificial Intelligence & Machine Learning Lab	
Course Outcome (CO)		**Bloom's Knowledge Level (KL)**
At the end of course, the student will be able to understand		
CO1	Analyze datasets by applying Python programming, NumPy, and Pandas libraries for data manipulation, preprocessing, and exploratory data analysis.	K4
CO2	Evaluate regression and classification models using supervised machine learning algorithms and appropriate performance evaluation metrics.	K5
CO3	Analyze datasets using unsupervised learning algorithms.	K4
CO4	Analyze dimensionality reduction techniques to discover hidden patterns and extract meaningful insights.	K4
CO5	Design and develop Artificial Neural Network models using TensorFlow/Keras for solving real-world predictive and classification problems	K6

Detailed Syllabus:

Experiment No.	Experiment
1	Familiarize students with Python programming fundamentals and Jupyter Notebook.
2	Perform numerical computations using the NumPy library.
3	Perform data loading, preprocessing, and manipulation using Pandas.
4	Visualize datasets using Matplotlib and Seaborn.
5	Prepare datasets for machine learning applications.
6	Implement and evaluate a Linear Regression model using Scikit-learn.
7	Develop a binary classification model using Logistic Regression.
8	Build and evaluate a Decision Tree classification model.
9	Implement the Random Forest algorithm for classification.
10	Implement the K-Nearest Neighbors algorithm for classification.
11	Implement a Support Vector Machine classifier.
12	Implement the Naïve Bayes algorithm for classification and evaluate its performance.
13	Perform unsupervised clustering using the K-Means algorithm.
14	Apply Principal Component Analysis for dimensionality reduction.
15	Develop a simple Artificial Neural Network using TensorFlow/Keras.

Note: The Instructor may add/delete/modify experiments, wherever he/she feels in a justified manner.